



Large Intestine & Anal Canal

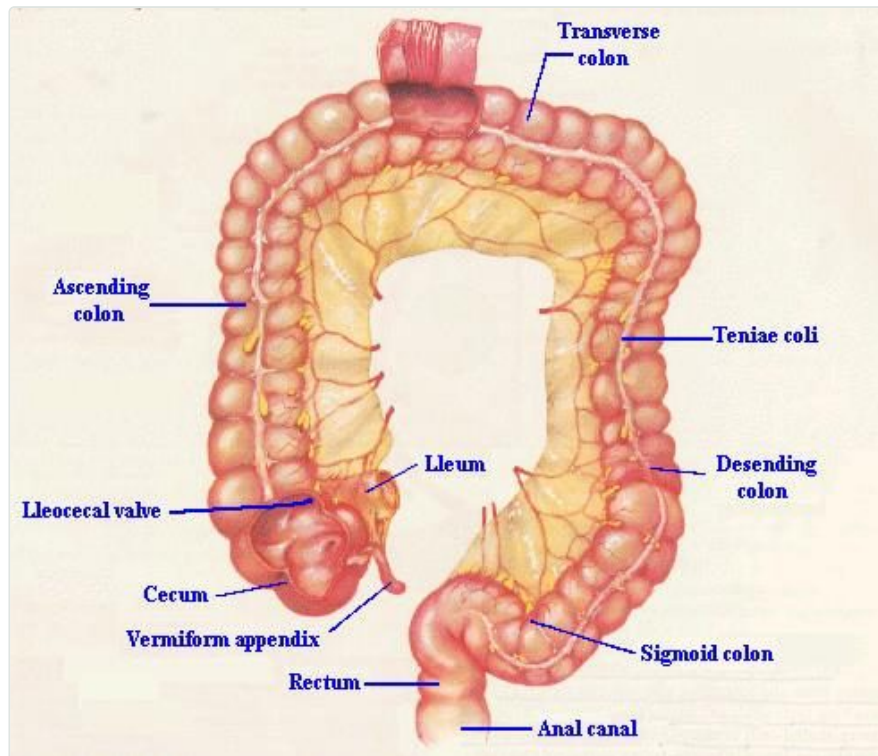
Histology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What 3 histologic features let you tell large intestine from small intestine at a glance?
2. Which region of the large intestine lacks teniae coli, and why does that matter?
3. What epithelial transition marks the dentate (pectinate) line?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Large intestine — regions



Ileocecal valve to anus: cecum → ascending, transverse, descending, sigmoid colon → rectum → anal canal.

Cecum & appendix	First part; appendix is a blind evagination of the cecum
Colon (~1.5 m)	Ascending → transverse → descending → sigmoid
Rectum	No teniae coli (see section 4)
Anal canal	Transition zone — its own section below

2 Large vs small intestine — the core discriminator

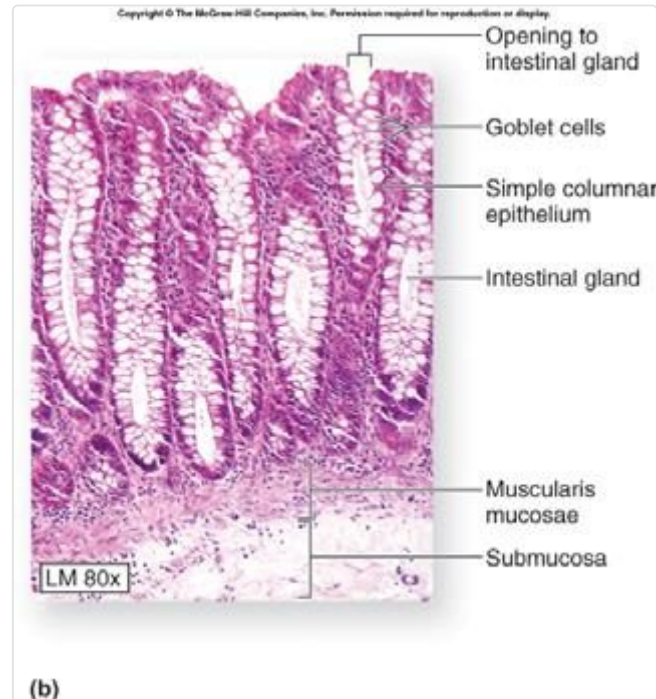
The single fastest ID call on an unlabelled slide.

Feature	Small intestine	Large intestine
Villi	Present	Absent
Folds	Plicae circulares (jejunum)	None (except rectal columns of Morgagni)
Crypts	Shorter	Long & straight Crypts of Lieberkuhn
Goblet cells	Moderate, increasing distally	Numerous
Lumen	Narrower	Wider , shorter tube overall
Lymphatic nodules	Peyer's patches — ileum only	Numerous throughout

Outer longitudinal muscle

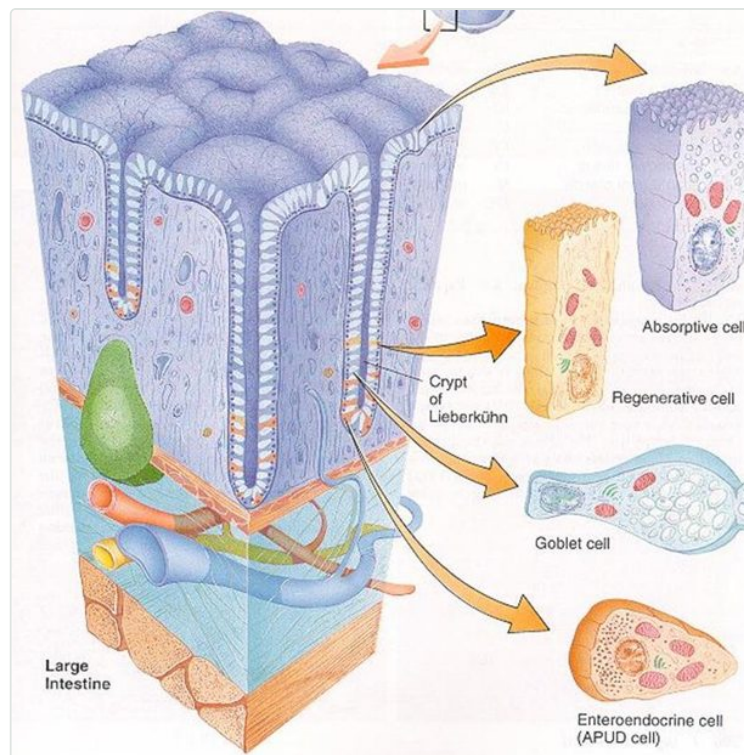
Continuous sheet

Gathered into 3 **teniae coli** (except appendix & rectum)



General colon architecture — goblet cells, simple columnar epithelium, intestinal glands, muscularis mucosae, submucosa.

3 Cells lining the mucosa & glands



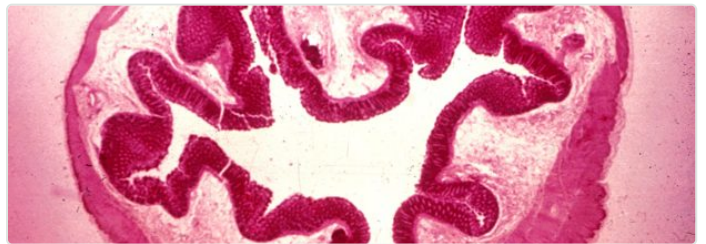
Absorptive, goblet, stem (regenerative) and enteroendocrine cells of the large-intestinal crypt.

Absorptive (columnar) cell	Most numerous surface cell
Goblet cell	Mucin granules — lubrication
Stem cell	Base of crypt; turns over in 3–4 days; precursor to other cells
Endocrine cell	Lower half of crypt; granules basal to the nucleus

Lecture did not mention Paneth or M cells for the large intestine — not included here to avoid overstating what was taught.

4 Within the large intestine — regional discriminators

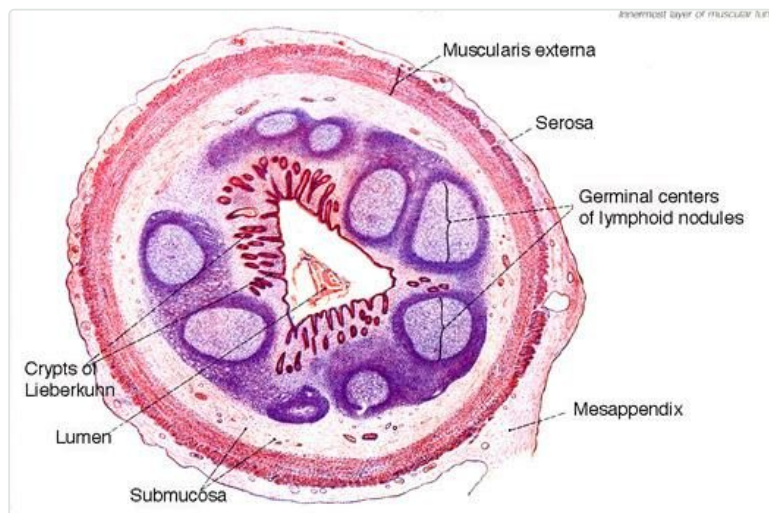
Region	Distinguishing histology
Cecum & colon	Standard picture above; teniae coli present . Serosa/adventitia depends on peritoneal relation (below)
Rectum	Same as colon but no teniae coli — outer longitudinal muscle is a complete sheet again; covered by adventitia
Appendix	Small, narrow, irregular lumen; abundant lymphoid nodules ; fewer & shorter glands than colon; no teniae coli ; muscularis externa is a normal 2-layer (inner circular + outer longitudinal) arrangement
Intraperitoneal	Cecum, transverse colon, sigmoid colon → covered by serosa
Retroperitoneal	Ascending & descending colon → adventitia + serosa
Rectum	Adventitia only



Teniae coli — thick longitudinal muscle bands, visible here in labelled and cross-sectional views.

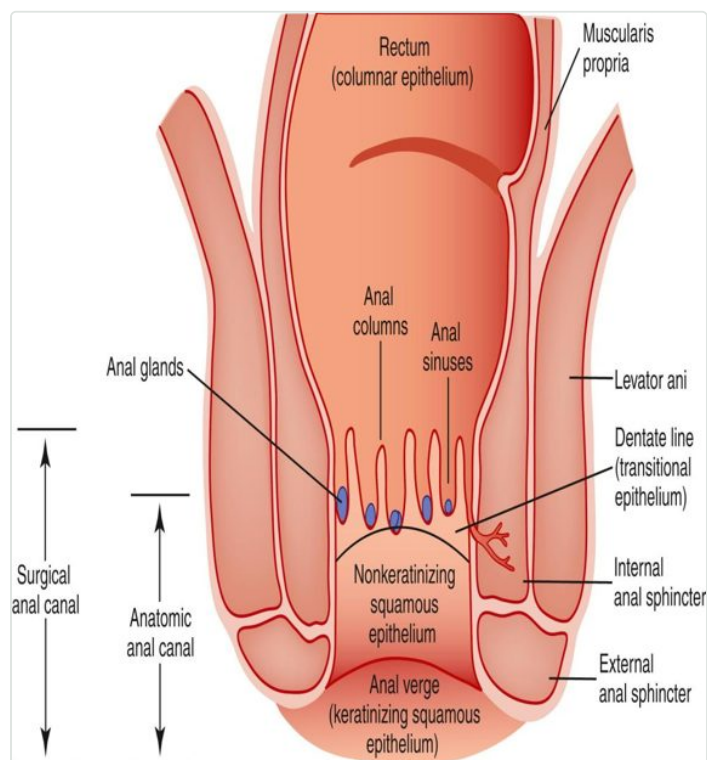


Rectum — same layers as colon, but no teniae coli.



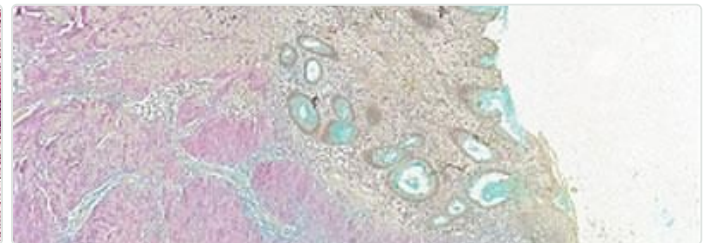
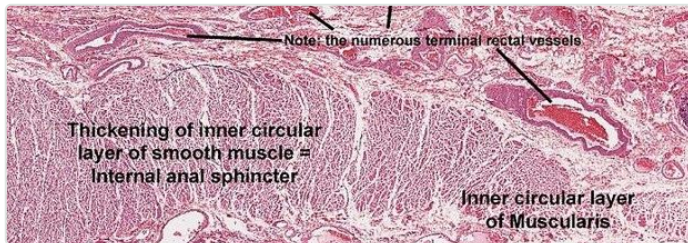
5 Anal canal — the zone that gets tested

~4 cm connecting rectum to the anal opening. Upper 3 cm = mucous membrane; lower 1 cm (8–10 mm) = true skin.



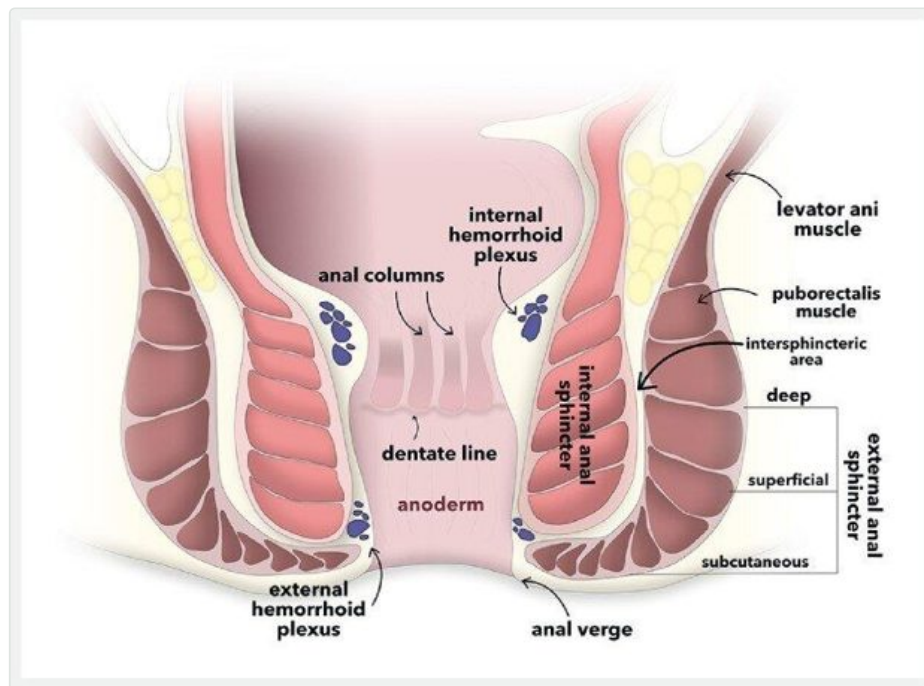
Zone	Approx. length	Epithelium	Landmark
Upper anal canal	15 mm	Simple columnar (like rectum)	Anal columns of Morgagni — 6–12 longitudinal folds
Lower part of upper 3 cm	15 mm	Non-keratinized stratified squamous	Anal valves join the column bases → together form the dentate (pectinate) line
Lowest anal canal	8–10 mm	True skin — keratinized stratified squamous	Hair follicles, sebaceous & sweat glands present

The dentate line marks the transition from simple columnar to stratified squamous epithelium, and divides the anal canal into an upper two-thirds and lower one-third.



Anorectal junction — columnar-to-squamous transition (left, H&E; right, van Gieson showing keratinized squamous epithelium of the anal canal vs crypts of Lieberkuhn of the rectum).

Venous plexus	Internal hemorrhoidal plexus — above the dentate line. External hemorrhoidal plexus — near the lower end of the canal
Internal anal sphincter	Thickened inner circular smooth muscle — involuntary , autonomic control
External anal sphincter	Outside the muscle layer — skeletal muscle , voluntary control



Internal vs external anal sphincter, relative to the dentate line and anal verge.

P Must-remember pearls

- Large intestine ID = **no villi** + long straight crypts + numerous goblet cells + abundant lymphoid nodules.
- **Teniae coli** = 3 bands of outer longitudinal muscle; **absent in rectum and appendix**.
- **Dentate (pectinate) line** = simple columnar → stratified squamous transition; splits the anal canal into upper 2/3 and lower 1/3.
- Lowest 8–10 mm of the anal canal = true skin — hair follicles, sebaceous & sweat glands present.
- Internal anal sphincter = smooth muscle, involuntary. External anal sphincter = skeletal muscle, voluntary.

Q Test yourself

1. A colonic biopsy shows long straight crypts, no villi, and abundant goblet cells. Which further feature would confirm this is from the transverse colon rather than the rectum?

- A) Presence of teniae coli
- B) Presence of Peyer's patches
- C) Presence of Brunner's glands
- D) Presence of villi

2. A biopsy from the appendix would be expected to show all of the following EXCEPT:

- A) Abundant lymphoid nodules
- B) Narrow, irregular lumen
- C) Teniae coli
- D) Fewer, shorter glands than colon

3. A transition from simple columnar epithelium to non-keratinized stratified squamous epithelium on histology marks which structure?

- A) Ileocecal valve
- B) Dentate line
- C) Recto-sigmoid junction
- D) Anal verge

4. The lowest 8-10 mm of the anal canal is histologically distinct because it is:

- A) Lined by simple columnar epithelium with Peyer's patches
- B) True skin with hair follicles and sweat glands
- C) Marked by teniae coli
- D) Lined by plicae circulares

5. Which anal sphincter is composed of skeletal muscle and under voluntary control?

- A) Internal anal sphincter
- B) External anal sphincter
- C) Both
- D) Neither

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. A — Teniae coli** — present in colon, absent in rectum; the fastest discriminator between the two.
- 2. C — Teniae coli** — the appendix has NO teniae coli despite otherwise resembling colon.
- 3. B — Dentate line** — marks the columnar-to-squamous switch and the upper 2/3 vs lower 1/3 boundary.
- 4. B — True skin** — hair follicles, sebaceous and sweat glands are present only in the lowest 8-10 mm.
- 5. B — External anal sphincter** — skeletal muscle, voluntary; internal sphincter is smooth muscle, involuntary.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Large Intestine and Anal Canal SGL3, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.